

PARH-OSSM: A Physiology-Aligned Harmonic State-Space Decomposition for Camera-Derived Respiratory Motion

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ABSTRACT

Camera-derived respiratory motion should not be forced into a single sinusoid. We study the current reproducible state of PARH-OSSM, a physiology-aligned harmonic state-space decomposition that separates respiratory motion into a second-order oscillatory component, a baseline trend, and an aperiodic residual. The audited implementation is an eight-dimensional exactly linear state-space model conditioned on an external frequency estimate, and it stores dual outputs for oscillatory analysis (z_{osc}) and full waveform analysis (z_{full}), together with causal and Rauch–Tung–Striebel smoothed trajectories and per-sample diagnostics.

This manuscript reports only claims supported by persistent artifacts in the current repository. A complete six-family COHFACE rerun is now available and is used for all quantitative tables. The MAHNOB-HCI production run is still incomplete, so no MAHNOB quantitative claims are made here. The live COHFACE ladder now includes raw OF, an additional OF-derived displacement bridge (OF_bridge), DoF, and three Profile1D families. On COHFACE, PARH-OSSM now exceeds KFstd in waveform fidelity on raw OF, is effectively tied on P1D-linear, slightly exceeds KFstd on P1D-quadratic in waveform CCC while remaining worse in waveform MAE/DTW, and still trails on OF_bridge, P1D-cubic, and DoF. In frequency tracking, PARH-OSSM improves over KFstd in five of the six families after routing rate metrics through the stored `track_hz`, but the raw base observation remains the strongest absolute rate route on raw OF, OF_bridge, and all three Profile1D families. Regenerated calibration diagnostics from the saved PARH-OSSM arrays yield median NIS means of roughly 0.83–0.94, median NIS in-band fractions of roughly 0.93–0.96, and Student- t weights that remain close to one, indicating conservative but internally consistent scaling rather than strong posterior downweighting. Mechanism audits from the saved per-frame diagnostics now show moderate dynamical-novelty scores, non-saturated oscillatory-support scores, and a family-dependent residual-energy share, indicating that the current implementation is no longer a nearly dormant decomposition scaffold.

We therefore position the current system as a physiology-aligned, self-auditing scaffold whose decomposition and traceability are already useful, but whose main bottleneck is still the observation model. The repository now contains multiple gated observation-calibration prototypes, and the current evidence supports two promotable observation-construction steps: a family-selective harmonic-only observation row for P1D-quadratic and P1D-cubic, and an additional OF-derived displacement-compatible family (OF_bridge) that improves OF-family rate tracking. Additional broad calibration variants, helper-only, helper-blend, and aggressive OF output-side policies all failed promotion.

Introduction

Video-based respiratory monitoring offers a contactless alternative to belts, thermistors, and capnography in settings where unobtrusive sensing is valuable, including sleep analysis, neonatal monitoring, and general bedside observation^{1–3}. Most video pipelines, however, still collapse the task to scalar rate estimation: a motion proxy is bandpass filtered and a dominant frequency is extracted by spectral or autocorrelation analysis^{4–6}. That framing is often adequate for stable recordings, but it is too restrictive for waveform morphology, baseline drift, and irregular events.

Two adjacent literature blocks motivate the present reset. First, state-space methods are only as meaningful as the correspondence between the chosen latent state, the observation equation, and the structure of the system that is believed to generate the data^{7–9}. That principle is especially important here because camera-derived respiration is not measured directly; it is observed through projection, extractor-family choice, and preprocessing. Recent benchmark work on RGB-camera respiration also shows that motion-based families, rPPG-based families, and learned pipelines behave very differently across datasets and evaluation targets¹⁰. Second, robust and adaptive filtering methods improve state estimation under outliers or changing noise statistics, for example through covariance adaptation, heavy-tailed likelihoods, or both^{11–15}. Third, oscillator state-space models

provide a principled way to represent quasi-periodic latent structure while preserving tractable inference under conditional linearity^{9,16–18}. Those lines of work are relevant but not sufficient on their own for camera-derived respiration, because they do not by themselves specify how respiratory morphology, baseline drift, and non-oscillatory event structure should be separated in the latent representation. Learned monitoring approaches can also be effective^{19,20}, but they do not remove the need for an interpretable model of what portion of the motion trace should be treated as oscillatory versus aperiodic.

The central premise of this work is that camera-derived respiratory motion is not a single clean sinusoid. It contains at least four interpretable parts: an oscillatory respiratory drive, harmonic asymmetry within each breath, baseline or projection drift, and residual structure that is respiratory but not well described by a narrow-band oscillator. A physiology-aligned model should separate those roles explicitly instead of forcing all mismatch into one innovation-magnitude logic. This premise is also consistent with respiratory physiology literature, which treats breathing as a variable pattern shaped by rhythm generation, mechanical feedback, non-respiratory motor acts, and behavioral or emotional modulation rather than as a perfectly stationary sinusoid^{21,22}.

This manuscript intentionally reports the *current verified state* of PARH-OSSM rather than an aspirational final system. A reverse-audit of the repository, saved results, and the previous manuscript showed that the older NAROSSM framing overclaimed what the present code and result files support. We therefore reset the paper around persistent evidence only.

The present manuscript makes three contributions.

- It defines the current minimal PARH-OSSM v1 as an eight-dimensional harmonic decomposition with dual outputs, exact linear filtering, and stored per-sample diagnostics.
- It reports the current reproducible evidence boundary: a completed COHFACE production run, saved diagnostic PKLs, and an incomplete MAHNOB production run that is not yet used for benchmark claims.
- It explains why corrected rate evaluation reveals a real PARH-OSSM rate gain over KFstd on COHFACE, while waveform gains remain limited despite the additional model structure.

The manuscript therefore proceeds in three layers. It first defines the code-verified model structure and its intended metric routing. It then reports only those quantitative results that are traceable to persistent artifacts. Finally, it interprets those artifacts mechanistically, including why the current COHFACE behavior is informative but still insufficient to stabilise the final paper story without rerun evidence from the updated code and from MAHNOB-HCI.

PARH-OSSM v1

Latent decomposition

The current audited model uses the latent state

$$\mathbf{x}_t = \left[h_{c,t}^{(1)}, h_{s,t}^{(1)}, h_{c,t}^{(2)}, h_{s,t}^{(2)}, b_t, \dot{b}_t, r_t, \dot{r}_t \right]^\top, \quad (1)$$

where $(h^{(1)}, h^{(2)})$ are the fundamental and second-harmonic oscillatory components, $(b_t, \dot{b}_t)$ is a slow baseline block, and $(r_t, \dot{r}_t)$ is an aperiodic residual block. The observation equation is

$$y_t = H\mathbf{x}_t + v_t, \quad H = [1, 0, 1, 0, 1, 0, 1, 0]. \quad (2)$$

The harmonic blocks use rotation-matrix dynamics conditioned on an external frequency estimate, the baseline block follows a constant-velocity model, and the residual block follows a damped random walk. Because the model remains linear conditional on the external frequency, standard Kalman filtering and Rauch–Tung–Striebel smoothing remain valid^{9,23,24}.

Dual outputs

The current implementation stores two outputs:

$$z_{\text{osc},t} = h_{c,t}^{(1)} + h_{c,t}^{(2)}, \quad z_{\text{full},t} = z_{\text{osc},t} + b_t + r_t. \quad (3)$$

z_{osc} is the intended oscillatory output for rate and dominant rhythm analysis. z_{full} is the intended full respiratory motion estimate for waveform fidelity and morphology. Both causal and smoothed versions are stored in the saved trial payloads.

Why this decomposition is physiology-aligned

The model structure is not arbitrary. A second harmonic is included because respiratory motion is usually asymmetric over each breath cycle even when the dominant rate is stable; forcing that asymmetry into a purely sinusoidal state invites harmonic confusion and pushes morphology into innovation residuals. The baseline block is needed because camera-derived chest motion carries slow projection drift and posture-related movement that are not themselves the respiratory rhythm. The residual block

exists for a different reason: it is the state-space location where irregular respiratory content should live when the oscillator support weakens, including amplitude transients or breath-shape departures that are still respiratory but not well explained by the harmonic submodel.

This decomposition also clarifies the evaluation policy. Rate metrics should be computed from z_{osc} , because rate is an oscillatory quantity. Waveform-fidelity metrics should be computed from z_{full} , because the full motion trace is the object of comparison when baseline and aperiodic content matter. That separation is one of the most important paper-level distinctions from the earlier manuscript state, where rate and waveform interpretations were more easily conflated. It also follows directly from the state-space design principle above: once the state variables are assigned distinct physiological roles, the observation targets and evaluation metrics must remain aligned with those roles rather than collapsing everything back to a single generic output.

Disentangled adaptation in the current code

The current scaffold separates three internal scores:

$$q_{\text{obs},t} \rightarrow \pi_t, \quad q_{\text{dyn},t} \rightarrow Q_{\text{osc},t}, \quad 1 - q_{\text{osc},t} \rightarrow Q_{\text{aper},t}. \quad (4)$$

Here π_t is a prior trust term used through $R_{\text{eff}} = R_t / \pi_t$, while Student- t downweighting remains a posterior safeguard through λ_t . In the current code, Mehra-style adaptive measurement noise remains the primary scale-setting mechanism¹¹; the quality scores modulate the prior trust and process noise around that baseline.

Current implementation boundary

The present repository already contains the following verified features: the eight-dimensional state, $K = 2$ harmonic structure, baseline and residual blocks, dual outputs, warm-up-only frequency initialisation, external helper frequency updates, causal and smoothed outputs, and saved per-sample diagnostics for R_t , π_t , λ_t , v_t , $q_{\text{obs},t}$, $q_{\text{dyn},t}$, and $q_{\text{osc},t}$.

Several important items are still incomplete or unstable. First, the current live production path is still not a fully general observation-calibrated state-space model. Instead, the code now uses a family-selective hybrid: raw OF uses a light observation path, OF_bridge is added as an OF-derived displacement-compatible observation family, DoF remains on the legacy stack, PID-linear remains on the legacy stack with selective frequency rescue, and PID-quadratic/cubic now use a bounded warm-up observation-row calibration. Second, broader warm-up observation-calibration prototypes and helper-only or helper-blend observation-routing variants failed the same 12-trial COHFACE promotion gate. This means preprocessing-policy switching alone is no longer a credible primary redesign axis. Third, the calibration CSV now emits NIS and lambda aggregates for PARH-OSSM from saved diagnostics arrays, but comparable per-frame calibration fields are still unavailable for KFstd. Fourth, the all-family COHFACE rerun has now been completed under the current live default including OF_bridge, but the MAHNOB-HCI production run is still incomplete, so the present manuscript does not claim cross-dataset results.

Observation-family semantics in the live scaffold

The most important implementation lesson so far is that the observation family is part of the model, not just part of preprocessing. Table 1 therefore summarises the six active COHFACE families by their current construction, observation domain, and present PARH role. Two contrasts matter most. First, raw OF and OF_bridge are both derived from optical flow but they are not interchangeable: raw OF remains a helper-heavy velocity-like family, whereas OF_bridge is a displacement-compatible constructed family whose main present benefit is rate. Second, the Profile1D quadratic and cubic routes are not just “better smoothed inputs”; they are currently the waveform-primary harmonic families because their observation semantics align best with the shared latent decomposition. This is the core reviewer-facing point of the present redesign: the same PARH latent state is kept across families, but what is observed and how it should be interpreted is explicitly family-dependent.

Results

Evidence boundary

All quantitative claims in this manuscript are restricted to persistent files generated by the completed all-family COHFACE rerun in `results/20260408_cohface_prod_ofbridge_full`. The evaluation CSVs, table-ready inputs, mechanism audit, and case-study manifest were all regenerated from that run. The same repository also contains gate-tested negative controls and pair-fusion experiments, but those remain secondary or explicitly rejected design evidence rather than current production behavior. The MAHNOB-HCI production directory still does not contain a completed metrics directory and metadata file, so MAHNOB-HCI is discussed only as the intended irregular-regime benchmark for the next stage, not as a current quantitative result.

Table 1. Observation-family semantics and current PARH role map. Values are generated from `paper/tables_ready/T2_observation_family_map.csv`, itself derived from the current semantics registry plus the official COHFACE table-ready T3/T4 CSVs.

Family	Construction	Domain	Primary information	Current PARH role	PARH T3 MAE	PARH T4 CCC
OF	raw vertical optical-flow surrogate	velocity	oscillatory motion	helper-heavy rate family	0.510	0.791
OF_bridge	OF-derived displacement-compatible bridge	displacement	bridged oscillatory displacement	rate-oriented constructed family	0.295	0.777
P1D_lin	linear 1D profile shift surrogate	displacement	fundamental displacement	conservative displacement family	0.465	0.727
P1D_quad	quadratic 1D profile shift surrogate	displacement	harmonic morphology	waveform-primary harmonic family	0.295	0.855
P1D_cub	cubic 1D profile shift surrogate	displacement	harmonic morphology	waveform-primary harmonic family	0.300	0.849
DoF	thresholded frame-difference count	energy	thresholded motion energy	nuisance-limited auxiliary family	2.110	0.571

COHFACE rate accuracy

Table 2 reports the current median per-trial COHFACE rate metrics after correcting the rate-evaluation path for methods that save `track_hz`. This materially changes the rate interpretation relative to the older legacy spectral path. PARH-OSSM now improves on KFstd in median MAE and RMSE in five of the six families, and it is the strongest estimator on DoF. The newly promoted `OF_bridge` route is also now the best PARH family by the locked T3 ladder rule, but even there the raw base and KFstd bridge routes still achieve lower absolute median error. The remaining limitation is different: the raw base observation still has the best median MAE on raw OF, `OF_bridge`, and all three Profile1D families.

This means the old “PARH $\approx$ KFstd” reading was partly an evaluation-path artifact, not a faithful description of the saved PARH-OSSM outputs. Once T3 is routed through the intended oscillatory output, PARH-OSSM is rate-competitive. The unresolved issue is that this rate improvement still does not translate into leading waveform fidelity. Figure 1 makes the rate-versus-family split explicit, especially the new raw-OF-versus-`OF_bridge` contrast.

Table 2. COHFACE frequency-domain rate accuracy. Values are median per-trial metrics from `paper/tables_ready/T3_rate_main.csv`, regenerated from `metrics_freq_domain_raw.csv`. Base uses the signal-spectral fallback; KFstd and PARH-OSSM use the stored `track_hz`.

Family	MAE (BPM)			RMSE (BPM)			Pearson r		
	Base	KFstd	PARH	Base	KFstd	PARH	Base	KFstd	PARH
OF	0.290	0.550	0.510	0.390	0.650	0.565	0.890	0.730	0.740
OF_bridge	0.245	0.260	0.295	0.325	0.335	0.370	0.900	0.835	0.850
P1D_lin	0.335	0.620	0.465	0.435	0.820	0.590	0.840	0.655	0.715
P1D_quad	0.195	0.485	0.295	0.270	0.580	0.375	0.950	0.800	0.890
P1D_cub	0.220	0.500	0.310	0.275	0.605	0.385	0.940	0.790	0.875
DoF	2.400	3.110	2.110	4.015	3.230	2.310	0.370	0.300	0.390

COHFACE waveform fidelity

Table 3 shows the current waveform comparison under the locked policy `Base(signal_hat) / KFstd(signal_hat) / PARH(z_full)`, all evaluated in smoothed mode. This is the strongest current positive result for PARH-OSSM: waveform fidelity improves over the raw base observation on every family except `OF_bridge`, where KFstd and PARH are already operating on a constructed OF-derived observation. However, the family-specific picture is mixed in a scientifically useful way: PARH-OSSM now exceeds KFstd on raw OF, is essentially tied with a slight edge on P1D-linear, slightly exceeds KFstd on P1D-quadratic in waveform CCC while remaining worse in waveform MAE/DTW, and remains below KFstd on `OF_bridge`, P1D-cubic, and DoF.

This pattern is materially different from a superiority claim. The current evidence supports “useful waveform improvement over the raw proxy” and “family-dependent but still incomplete narrowing of the KFstd gap”.

Figure 2 makes the current best waveform family more concrete by holding the trial fixed and comparing Base, KFstd, and PARH on the same `P1D_quad` clips. This same-trial view matters because the family-level medians in Table 3 can hide where the current PARH route is genuinely stronger and where it still trails KFstd. In the best and median clips, PARH narrows or matches the KFstd waveform shape while maintaining the physiology-aligned decomposition. In the bottom clip, however, the figure also makes the current limitation explicit: once the observation is already badly mismatched, the present observation model does not rescue the waveform.

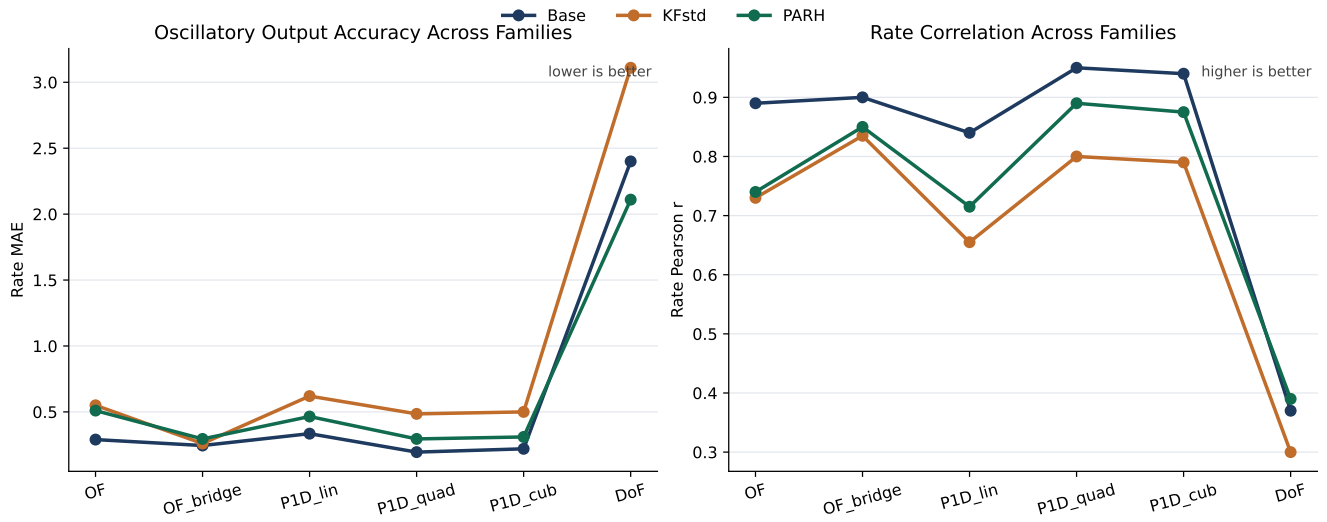


Figure 1. Family-level COHFACE T3 summary generated from `paper/tables_ready/T3_rate_main.csv`. The left panel shows median rate MAE and the right panel shows median Pearson correlation across the six active observation families. The figure makes the current split explicit: `OF_bridge` is the best PARH rate route, while raw base observations remain strongest in absolute rate error for several displacement-like families.

Table 3. COHFACE waveform fidelity. Values are median per-trial metrics from `paper/tables_ready/T4_waveform_main.csv`, regenerated from `metrics_waveform_raw.csv`.

Family	CCC			wMAE			DTW		
	Base	KFstd	PARH	Base	KFstd	PARH	Base	KFstd	PARH
OF	0.711	0.773	0.791	0.576	0.516	0.511	0.452	0.403	0.403
OF_bridge	0.685	0.789	0.777	0.629	0.503	0.504	0.466	0.396	0.399
P1D_lin	0.578	0.725	0.727	0.655	0.536	0.543	0.531	0.424	0.424
P1D_quad	0.811	0.853	0.854	0.484	0.416	0.425	0.368	0.343	0.346
P1D_cub	0.806	0.851	0.848	0.490	0.419	0.425	0.371	0.344	0.346
DoF	0.438	0.575	0.571	0.792	0.731	0.736	0.551	0.460	0.459

COHFACE calibration diagnostics

Table 4 summarises the regenerated PARH-OSSM calibration CSV. The current evaluation path now emits NIS and lambda aggregates directly from the saved PARH-OSSM diagnostic arrays, while KFstd remains unavailable in this table because the saved KFstd payloads do not contain analogous per-frame diagnostics.

Across families, PARH NIS means lie between 0.833 and 0.937 and NIS in-band fractions between 0.929 and 0.958. Lambda means remain close to one, and the fraction of downweighted frames is highest on raw OF at 8.6% while remaining near zero on `OF_bridge` and the profile families. `OF_bridge` is informative here: it lowers the OF-family NIS mean and raises the in-band fraction, which is consistent with a more observation-consistent rate-oriented route, but its stability still remains far below the profile families.

Table 4. COHFACE PARH-OSSM calibration diagnostics from `paper/tables_ready/T6_diagnostics_main.csv`. KFstd remains blank in this table because its saved payloads do not carry comparable per-frame diagnostic arrays.

Family	NIS mean	NIS in-band	λ mean	$\Pr(\lambda_t < 1)$	Stability (s)
OF	0.937	0.929	1.019	0.086	9.825
OF_bridge	0.841	0.950	1.015	0.000	18.575
P1D_lin	0.839	0.942	1.017	0.005	60.450
P1D_quad	0.833	0.949	1.017	0.000	60.525
P1D_cub	0.835	0.951	1.016	0.000	60.500
DoF	0.891	0.958	1.009	0.014	40.500

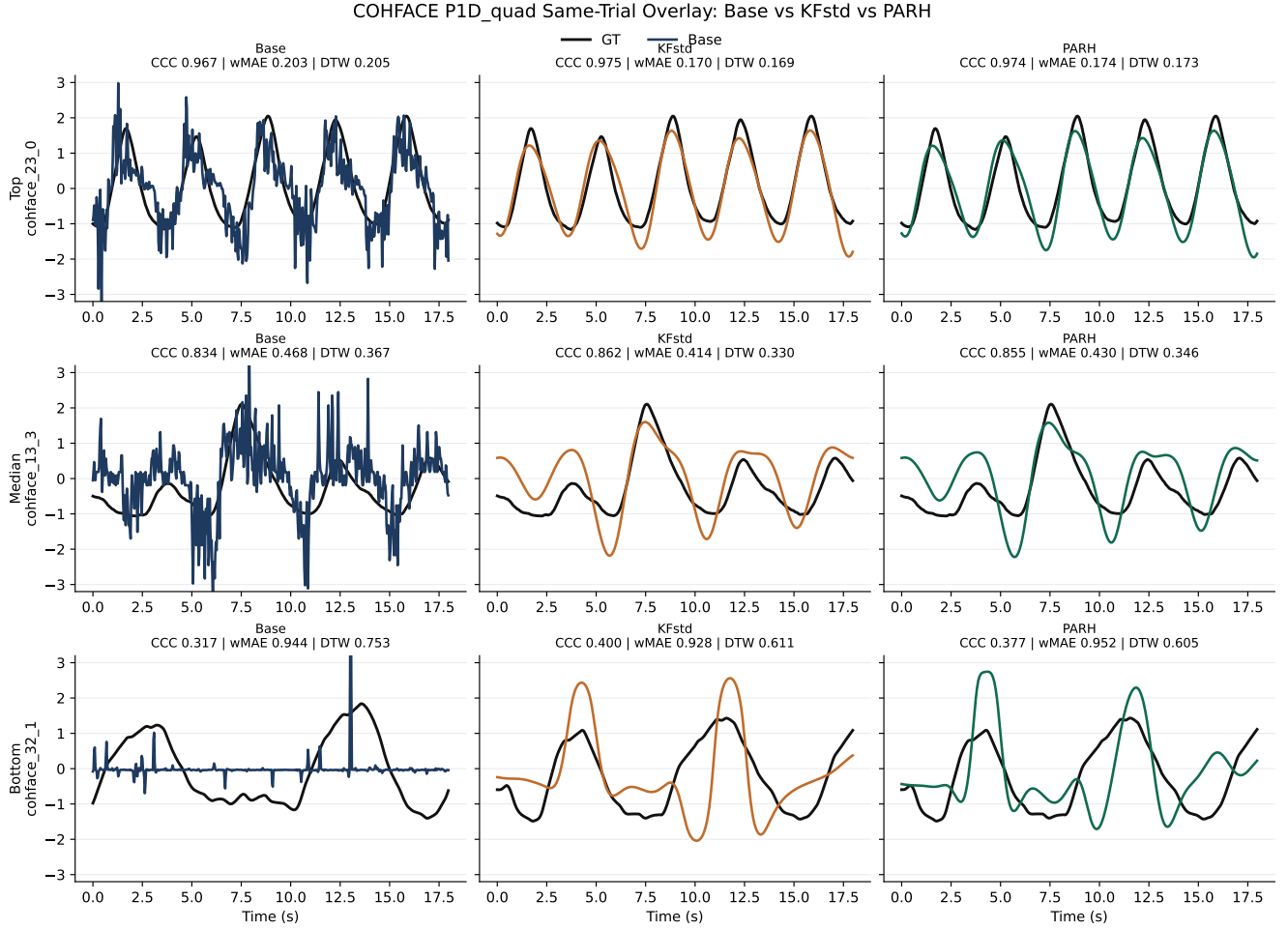


Figure 2. Same-trial waveform overlays for the current strongest waveform family P1D_quad. Rows are the top, median, and bottom PARH clips selected by waveform_CCC using the persistent manifest `paper/manifests/cohface_pldquad_overlay_manifest.csv`. Columns show the corresponding Base, KFstd, and PARH outputs on the same trial, each aligned to the ground-truth respiratory waveform using the same cross-correlation rule as the T4 protocol.

COHFACE mechanism audit

To understand why PARH-OSSM now shows better T3 rate metrics than KFstd while still falling short of KFstd on waveform fidelity, we aggregated the saved per-sample diagnostics directly from the persistent trial PKLs of the completed production run. Table 5 summarises the resulting mechanism profile.

The pattern is now materially healthier than in the older conservative scaffold. Observation trust remains only mildly reduced on average ($q_{\text{obs}} \approx 0.91\text{--}0.93$). The dynamical-novelty score is no longer near-saturated: q_{dyn} lies around 0.56–0.69 across families, while q_{osc} now lies around 0.73–0.83 rather than near one. The aperiodic pathway therefore opens more often than before, but its activity is clearly family-dependent. Raw OF, OF_bridge, PID-linear, and DoF retain non-zero residual energy, whereas the stronger PID-quadratic/cubic routes now drive almost all clean COHFACE structure through the harmonic block. Strong Student- t downweighting is still modest overall, but raw OF remains the family with the clearest low-level tail fraction.

In other words, the current implementation is no longer a nearly dormant extension of KFstd. The extra decomposition states are being used. The remaining limitation is that this healthier decomposition still does not close the family-specific observation gap strongly enough to dominate the best raw base proxies. Figure 3 shows this visually: the mechanism scores are active and calibrated, but the family-level energy routing is still uneven.

Table 5. COHFACE mechanism audit from saved PARH-OSSM diagnostic arrays in the production PKLs. Values are pooled means unless otherwise noted.

Family	q_{obs}	q_{dyn}	q_{osc}	$\Pr(\lambda_t < 0.95)$	$Q_{\text{osc}}/Q_{\text{osc},0}$	$Q_{\text{aper}}/Q_{\text{aper},0}$	E_b/E_{tot}	E_r/E_{tot}
OF	0.911	0.594	0.831	0.055	1.297	1.677	0.023	0.011
OF_bridge	0.925	0.554	0.788	0.006	1.277	1.847	0.026	0.013
P1D_lin	0.926	0.588	0.788	0.021	1.294	1.848	0.021	0.012
P1D_quad	0.928	0.563	0.787	0.007	1.281	1.850	0.000	0.000
P1D_cub	0.928	0.564	0.786	0.008	1.282	1.857	0.000	0.000
DoF	0.929	0.692	0.727	0.007	1.346	2.094	0.011	0.018

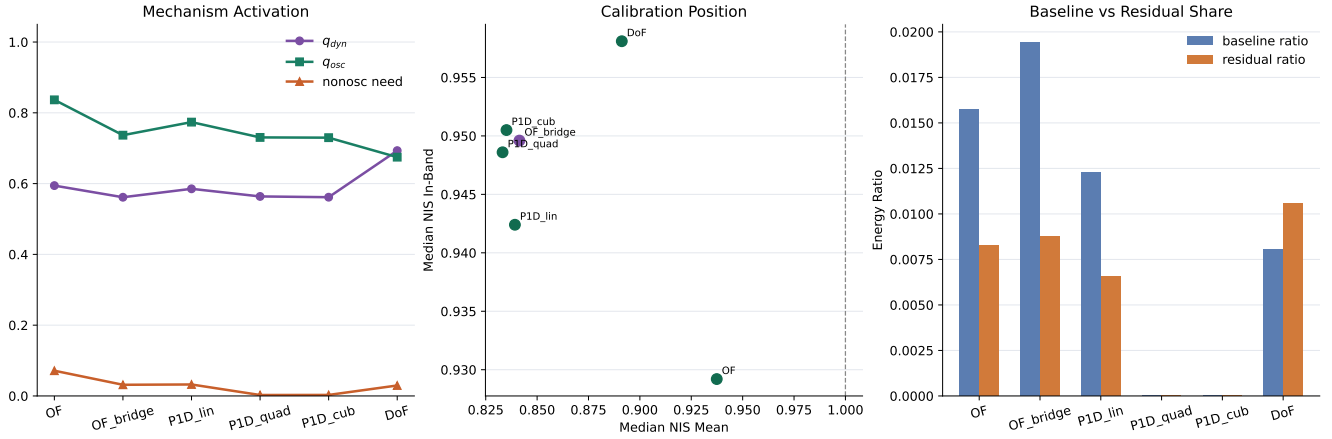


Figure 3. Mechanism and calibration summary from the current six-family COHFACE rerun. Left: family-level median mechanism activity, showing moderate q_{dyn} , non-saturated q_{osc} , and a small but non-zero observation-driven non-oscillatory need. Middle: family-level PARH calibration position in NIS mean versus NIS in-band space. Right: baseline and residual energy ratios, showing that residual usage remains family-dependent and near-zero on the strongest quadratic/cubic profile routes.

Discussion

Why current PARH-OSSM gains are limited on COHFACE

After correcting the T3 evaluation path, COHFACE no longer supports the claim that PARH-OSSM is merely tied with KFstd. The current saved PARH-OSSM outputs are better than KFstd for rate tracking in five of the six observation families. The remaining limitation is narrower but still important: PARH-OSSM still fails to beat the raw base observation on rate MAE in every family except DoF, and its waveform gains remain family-specific rather than uniform.

The main bottleneck is now better described as an observation-model problem than as a dormant-latent problem. The current scaffold no longer shows the old near-saturated q_{osc} behavior. Instead, the saved diagnostics show moderate q_{dyn} , non-saturated q_{osc} , and residual energy shares of roughly 0.7–1.8%. The decomposition is therefore active. What remains unresolved is that the observation equation still does not encode family-specific visibility, mild lag mismatch, or nuisance projection effects well enough to consistently outperform the strongest raw proxies.

The negative gate results point in the same direction. Exposing helper-style observations to quad/cub families slightly improves rate but slightly harms waveform, helper routing for all P1D families clearly breaks P1D-linear, and helper-blend policies are slightly worse than the current scaffold. That means continued preprocessing-policy switching is not the answer. The next gain has to come from stronger observation-row semantics rather than from another preprocessing rearrangement. The current evidence now supports that claim in two complementary ways: a selective harmonic-only observation row helps the two higher-order profile families, and an OF-derived displacement bridge changes the OF-family rate/waveform trade-off in a reproducible direction. The resulting best-single ladder is therefore split by task: OF_bridge is now the strongest PARH T3 family under the locked median-based ladder rule, whereas P1D_quadratic remains the strongest PARH T4 family.

What already matches the intended final model

The current implementation is not a placeholder in the trivial sense. The core architecture already matches the intended PARH-OSSM direction in several important ways: the eight-dimensional latent decomposition is real, the second-harmonic morphology block is active, the baseline and residual states exist in the saved trajectories, warm-up-only initialisation is

implemented, the forward and RTS-smoothed paths are separated, and the saved result payloads already contain the diagnostics needed for mechanism-level auditing.

That matters because the current issue is not “the model was never built”. The issue is that the presently implemented adaptation maps and evaluation path do not yet exploit that structure in an intent-aligned way.

What remains approximate or wrong

Several mismatches still separate the current code from the intended final PARH-OSSM story.

- The current COHFACE rerun now reflects the separated helper path, softer $q_{\text{dyn}}/q_{\text{osc}}$ mappings, corrected Student- t algebra, and warm-up-only KFstd initialisation, but it still does not solve the observation bottleneck.
- The calibration CSV is now informative for PARH-OSSM, but comparator symmetry is incomplete because KFstd does not save matching per-frame calibration arrays and posterior frequency uncertainty is still missing.
- Three observation-calibration prototypes now exist in the repository, but the component-wise, global-scalar, and oscillator-vs-auxiliary variants have all failed to produce a promotable COHFACE gate result.
- Additional helper-only and helper-blend observation-routing variants also failed the same gate, which means the next redesign step must move beyond preprocessing-policy switching.
- The current bridge observation path is still an interim family-specific workaround rather than a final observation-calibrated generative model.

Limitations and next steps

This manuscript intentionally omits several claims that were present in the previous draft. There is no MAHNOB-HCI quantitative section because the current production run was incomplete at audit time. The current T6 table is PARH-specific because KFstd does not yet expose matching per-frame diagnostics. There are no cross-dataset transfer or synthetic outlier-injection claims because the current persistent production artifacts do not support them.

The repository now contains a live COHFACE scaffold reference, so the next stage should prioritise four concrete tasks before the manuscript is stabilised: (1) retain the six-family COHFACE ladder as the current evidence boundary; (2) strengthen OF-family observation semantics beyond the present raw-vs-bridge split while preserving waveform meaning; (3) make the residual branch more identifiable than the current release heuristics allow; and (4) complete the MAHNOB-HCI production run under the current code while extending comparator-side diagnostics so calibration and uncertainty can be compared more symmetrically. Only then will it be possible to assess whether the decomposition truly pays off in irregular regimes rather than only in stationary ones.

Methods

Observation families

We use six heterogeneous chest-motion proxy families that are already implemented in the repository: raw OF-Farneback dense optical flow, an OF-derived displacement-compatible bridge (OF_bridge), DoF, and three one-dimensional spatial profile families (linear, quadratic, and cubic). These observation families are treated as separate single-family inference paths rather than fused into one multi-family model in the current manuscript. That restriction is intentional: the current evidence boundary supports fair single-family comparison, not yet a joint-fusion claim.

These families should not be described as if they all observe the same waveform. Raw OF-Farneback is the median vertical component of dense optical flow between adjacent ROI frames, so it is closer to a signed oscillatory motion or velocity-like proxy than to absolute displacement. OF_bridge is derived from raw OF through an OF-to-displacement bridge and therefore acts as a displacement-compatible OF-family route rather than a new extractor. DoF is a thresholded count of large inter-frame pixel changes and is therefore closer to motion energy than to a clean respiratory waveform. The Profile1D families estimate a frame-to-frame chest-shift surrogate from one-dimensional vertical profiles and are therefore closer to displacement-like proxies, with the quadratic and cubic variants exposing richer harmonic morphology than the linear variant. The observation model should therefore use a shared latent respiratory decomposition but family-specific observation semantics.

This also explains why the current redesign introduces an additional OF_bridge family but does not yet introduce an analogous “DoF-bridge.” Raw OF retains signed directional structure from a motion field, which makes a displacement-compatible reconstruction at least identifiable in principle. DoF does not: it is an aggregated motion-energy count after thresholding, so its sign, phase, and direct displacement meaning are already collapsed before state-space inference begins. Under the present code and data, DoF is therefore better treated as a nuisance-limited auxiliary family than as a near-displacement waveform proxy that merely awaits a bridge transform.

For that reason, preprocessing is part of the observation definition rather than a generic denoising block. The raw extractor output, the current inference-path signal, and the helper-path signal all have different meanings. The current scaffold now

reflects this more explicitly: raw OF uses a light observation path, OF_bridge provides a separate OF-derived displacement-compatible route, DoF remains on the legacy path because nuisance contamination is still too strong, PID-linear remains a conservative displacement-like family with selective frequency rescue, and PID-quadratic/cubic are the only families for which bounded observation-side calibration has passed the COHFACE no-harm gate. Stronger relaxed OF helper blending was also tested but failed the full-dataset confirmation step, which reinforces the need for better OF observation semantics rather than a looser output-side gate. The intended final model keeps the latent state shared across families while replacing the current generic observation row with bounded family-specific visibility and lag semantics. The current COHFACE evidence also shows why constructed observation families deserve to be treated as first-class model objects: raw OF and OF_bridge invert the usual rate-versus-waveform ranking across the same underlying optical-flow source, which would be impossible to explain if observation construction were merely a cosmetic preprocessing step.

Figure 4 makes two of these limitations visible without post hoc cherry-picking because both panels are selected from persistent manifests. The top row shows a residual-heavy raw-OF PARH case. The residual branch is clearly active and the non-oscillatory-need trace rises, but the waveform fit remains poor. This is exactly the current repository truth: the residual path is no longer dormant, yet it is still not a fully identifiable physiological branch that can rescue a strongly mismatched observation. The bottom row shows the worst OF_bridge case from the bridge manifest. This keeps the story honest: OF_bridge is a useful new observation family for rate-oriented analysis, but it is not a universal replacement for raw OF. That is why the paper currently promotes OF_bridge as an additional family rather than collapsing the two.

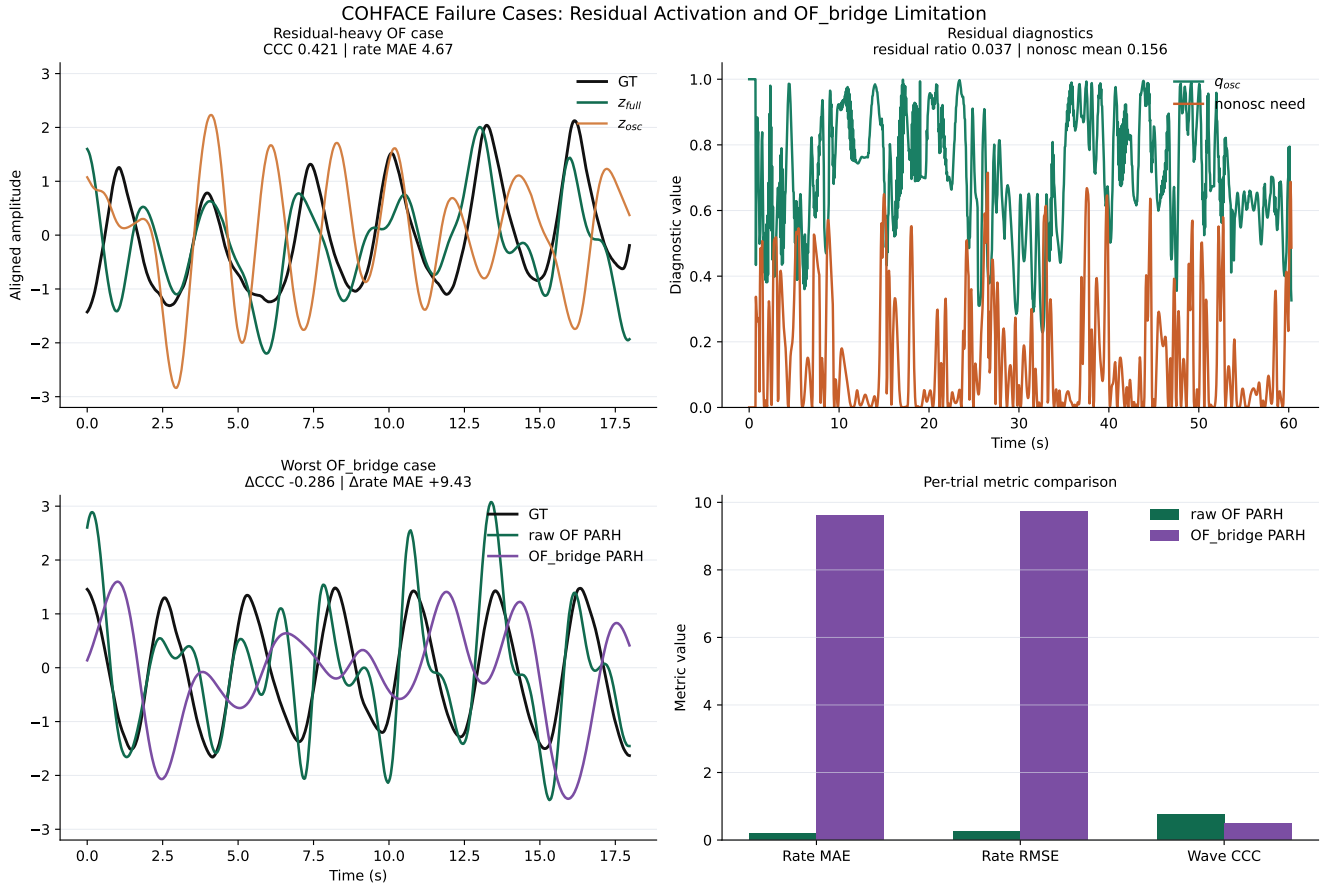


Figure 4. Failure cases selected from persistent manifests. Top row: COHFACE_residual_heavy_of from paper/manifests/cohface_failure_case_manifest.csv shows a raw-OF PARH trial with elevated residual energy and non-oscillatory need, but still poor waveform fidelity. Bottom row: the worst OF_bridge case from the same manifest shows that observation construction is not uniformly beneficial. Together these panels bound the current claims: the model is more interpretable than a generic robust filter, but the observation and residual branches are still incomplete.

Datasets and current artifact status

COHFACE contains 160 recordings from 40 subjects in a controlled lab setting²⁵. MAHNOB-HCI provides a more heterogeneous emotion-elicitation environment with longer and higher-frame-rate recordings²⁶. The intended final PARH-OSSM paper uses COHFACE as a regular, do-no-harm regime and MAHNOB-HCI as an irregular regime where the residual pathway should matter more.

At the time of the present manuscript state, the all-family COHFACE production run including OF_bridge had completed and its metrics and metadata were available. The MAHNOB-HCI production directory still lacks the completed metrics and metadata required for benchmark tables. We therefore restrict quantitative claims to COHFACE. Figure 5 shows why the observation family should be treated as part of the modelled measurement channel rather than as a generic interchangeable preprocessing choice.

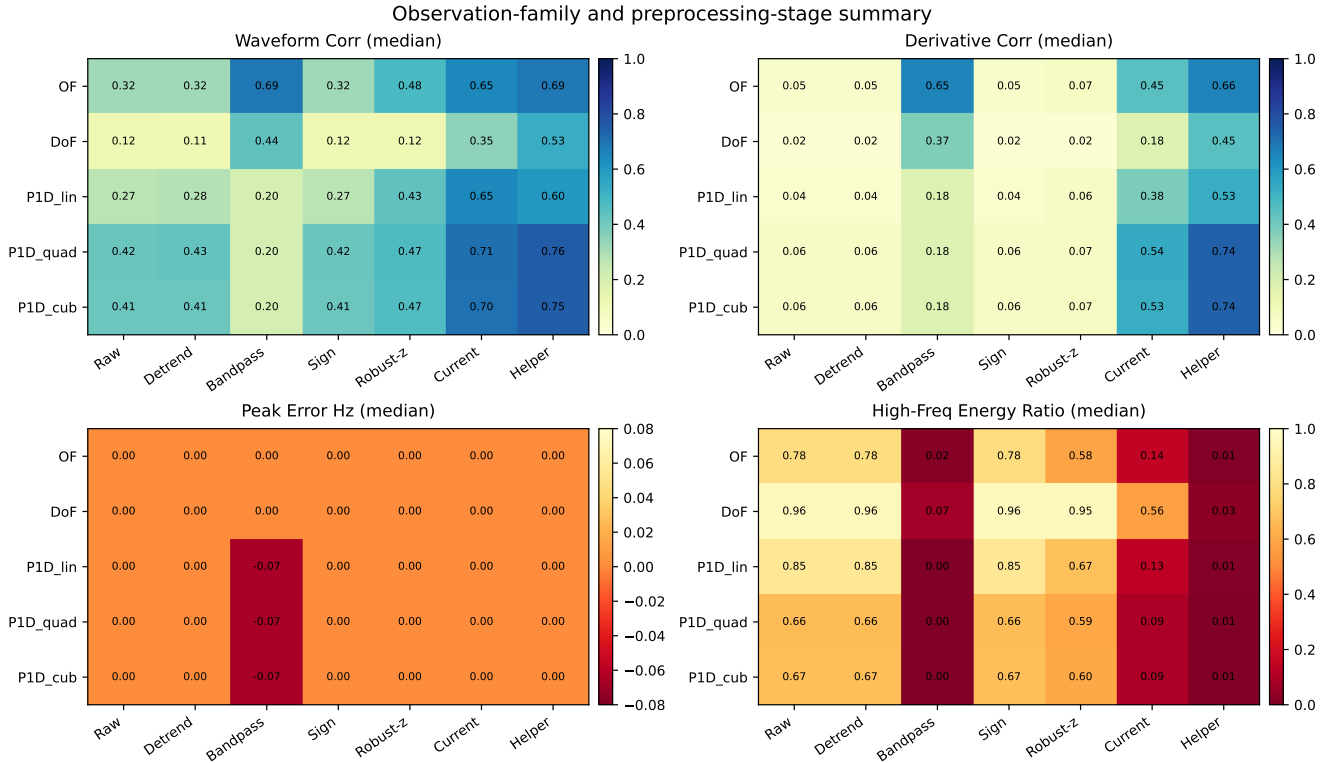


Figure 5. Dataset-regime and observation-characterization summary derived from the current COHFACE observation EDA. The family-dependent preprocessing effect is part of the scientific claim: the observation gap is not uniform across raw OF, OF_bridge, DoF, and the Profile1D families, so the final model must use family-specific observation semantics rather than one shared preprocessing assumption.

Evaluation protocol

The current manuscript uses three quantitative sources.

- Table 2 comes from the regenerated `paper/tables_ready/T3_rate_main.csv`, which is itself derived from the persistent `metrics_freq_domain_raw.csv` file. Base rows use the signal-spectral fallback; KFstd and PARH-OSSM use the saved `track_hz`.
- Table 3 comes from the regenerated `paper/tables_ready/T4_waveform_main.csv`, which is derived from `metrics_waveform_raw.csv`. Under the current locked policy, Base and KFstd use `signal_hat` while PARH-OSSM uses `z_full`, all in smoothed mode.
- Table 4 comes from the regenerated `paper/tables_ready/T6_diagnostics_main.csv`, which is derived from `metrics_filter_diagnostics_raw.csv`. In the current repository state, meaningful NIS/lambda rows are available only for PARH-OSSM because those arrays are saved in the PARH trial payloads.

This differs intentionally from the evaluation protocol adopted in the `resPyre` benchmark review¹⁰. In that benchmark, each method is first treated as a respiratory-waveform estimator and respiration rate is then derived from the estimated waveform

by spectral peak selection. That design is appropriate for comparing heterogeneous extractors in a common benchmark pipeline, but it is not sufficient for the current PARH question because it collapses oscillatory-rate behaviour and full-waveform behaviour back onto a single output. We therefore keep the benchmark spirit and datasets, but change the evaluation routing to match the latent semantics: T3 uses the oscillatory output when the model provides it, T4 uses the full output, and T6 reports calibration diagnostics separately.

The waveform metrics follow a common pipeline: bandpass filtering to the respiratory band, per-signal z-scoring, cross-correlation alignment, then whole-trial concordance correlation coefficient, waveform MAE, and normalised DTW²⁷. Calibration metrics are computed from saved per-frame diagnostics arrays when present, with frame logs used only for supplementary failure and uncertainty fields.

For the final submission, all overlay figures and case studies should be drawn from a persistent case-selection manifest rather than manual inspection, and all confirmatory method comparisons should use paired non-parametric tests on the regenerated rerun artifacts. That policy is not yet exercised in the current manuscript because the necessary MAHNOB-HCI rerun and ablation artifacts are still pending, but it is already locked at the paper-planning level to reduce reviewer exposure to cherry-picking or post hoc metric selection.

Implementation audit procedure

Beyond the table-ready CSVs, we audited the current implementation by reading the model, wrapper, and evaluation code paths and by aggregating the saved per-sample diagnostic arrays directly from the COHFACE production PKLs. This allowed us to compare the intended design against the actual implementation and to diagnose why the current COHFACE behaviour now shows family-specific gains: better rate tracking than KFstd in most families, an observation-construction gain for OF_bridge, and still incomplete waveform dominance in the stronger profile families.

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Author contributions statement

Y.-J.C. performed the reverse-audit, implemented the current PARH-OSSM code base, analysed the saved artifacts, and prepared the manuscript. D.-Y.K. supervised the project and reviewed the manuscript direction and interpretation.

Additional information

The authors declare no competing interests.